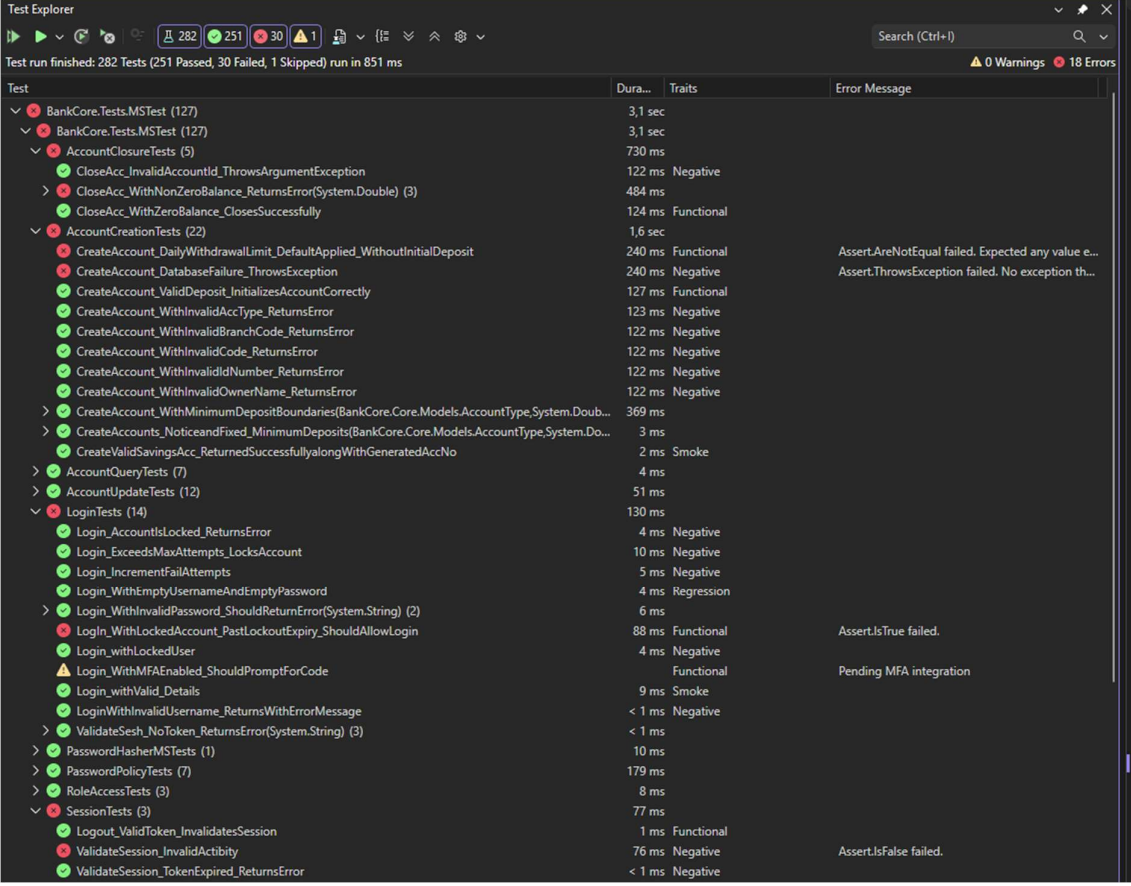


Automated test run screenshots:

NOTE: As per section 6.4, bug fixes were simulated as fixed, thus why not all my tests are green, because they are meant to identify the bugs. A true green state will be achievable only once the development team fixes the bugs identified within the automated test runs.



Test Explorer

Test run finished: 282 Tests (251 Passed, 30 Failed, 1 Skipped) run in 851 ms

0 Warnings 18 Errors

Test	Dura...	Traits	Error Message
BankCore.Tests.MSTest (127)	3,1 sec		
BankCore.Tests.MSTest (127)	3,1 sec		
AccountClosureTests (5)	730 ms		
CloseAcc_InvalidAccountId_ThrowsArgumentException	122 ms	Negative	
CloseAcc_WithNonZeroBalance_ReturnsError(System.Double) (3)	484 ms		
CloseAcc_WithZeroBalance_ClosesSuccessfully	124 ms	Functional	
AccountCreationTests (22)	1,6 sec		
CreateAccount_DailyWithdrawalLimit_DefaultApplied_WithoutInitialDeposit	240 ms	Functional	Assert.AreEqual failed. Expected any value e...
CreateAccount_DatabaseFailure_ThrowsException	240 ms	Negative	Assert.ThrowsException failed. No exception th...
CreateAccount_ValidDeposit_InitializesAccountCorrectly	127 ms	Functional	
CreateAccount_WithInvalidAccType_ReturnsError	123 ms	Negative	
CreateAccount_WithInvalidBranchCode_ReturnsError	122 ms	Negative	
CreateAccount_WithInvalidCode_ReturnsError	122 ms	Negative	
CreateAccount_WithInvalidIdNumber_ReturnsError	122 ms	Negative	
CreateAccount_WithInvalidOwnerName_ReturnsError	122 ms	Negative	
CreateAccount_WithMinimumDepositBoundaries(BankCore.Core.Models.AccountType,System.Doub...	369 ms		
CreateAccounts_NoticeandFixed_MinimumDeposits(BankCore.Core.Models.AccountType,System.Do...	3 ms		
CreateValidSavingsAcc_ReturnedSuccessfullyalongWithGeneratedAccNo	2 ms	Smoke	
AccountQueryTests (7)	4 ms		
AccountUpdateTests (12)	51 ms		
LoginTests (14)	130 ms		
Login_AccountIsLocked_ReturnsError	4 ms	Negative	
Login_ExceedsMaxAttempts_LocksAccount	10 ms	Negative	
Login_IncrementFailAttempts	5 ms	Negative	
Login_WithEmptyUsernameAndEmptyPassword	4 ms	Regression	
Login_WithInvalidPassword_ShouldReturnError(System.String) (2)	6 ms		
Login_WithLockedAccount_PastLockoutExpiry_ShouldAllowLogin	88 ms	Functional	Assert.IsTrue failed.
Login_withLockedUser	4 ms	Negative	
Login_WithMFAEnabled_ShouldPromptForCode		Functional	Pending MFA integration
Login_withValid_Details	9 ms	Smoke	
LoginWithInvalidUsername_ReturnsWithErrorMessage	< 1 ms	Negative	
ValidateSesh_NoToken_ReturnsError(System.String) (3)	< 1 ms		
PasswordHasherMSTests (1)	10 ms		
PasswordPolicyTests (7)	179 ms		
RoleAccessTests (3)	8 ms		
SessionTests (3)	77 ms		
Logout_ValidToken_InvalidatesSession	1 ms	Functional	
ValidateSession_InvalidActivity	76 ms	Negative	Assert.IsFalse failed.
ValidateSession_TokenExpired_ReturnsError	< 1 ms	Negative	

Test Explorer

Test run finished: 282 Tests (251 Passed, 30 Failed, 1 Skipped) run in 851 ms

0 Warnings 18 Errors

Test	Dura...	Traits	Error Message
BankCore.Tests.NUnit (70)	189 ms		
ReportingEngineNUnitTests (10)	116 ms		
AttemptingToGenerateStatement_WithFromDate_AfterToDate (2)	74 ms		
GetAuditLog_InvalidDateRange_ReturnsError (2)	2 ms		
GetAuditLog_ValidDateRange (2)	1 ms		
GeneratingAStatement_WithValidInfo	15 ms	Positive, Reporting Service	
GeneratingAValidSummaryReport	20 ms	Positive, Reporting Service	
UsingInvalidAccount_WhenGeneratingAStatement	4 ms	Negative, Reporting Service	
UsingPendingAccount_ToGenerateStatement	< 1 ms	Negative, Reporting Service	
TransactionEngineNUnitTests (60)	73 ms		
DecisionTable_WithdrawProcessTransaction (8)	7 ms		
MakeDeposit_InvalidAmounts_ReturnsError (3)	< 1 ms		
MakeDeposit_InvalidAndNegativeAmount_ReturnsError (2)	< 1 ms		
MakeTransfer_ValidAndInvalidScenarios (3)	2 ms		
PerformDeposit_CombinatorialRules (16)	< 1 ms		
ReverseTransaction_EmptyRefField_ReturnsError (2)	< 1 ms		
BatchDepositTests_MultipleValidAccts	19 ms	Functional, TransactionEngine	
CurrencyValidation_DepositCheck	1 ms	Functional, TransactionEngine	
MakeDeposit_TransactionDBOffline_ThrowsException	1 ms	Critical, TransactionEngine	
MakeDeposit_ValidActiveAccount_UpdatesBalanceAndLogsTransaction	3 ms	Functional, TransactionEngine	
MakeROTransfer_ReturnsError		Negative, TransactionEngine	
MakeTransfer_FromAcc_with_InsufficientFunds_ReturnsError		Negative, TransactionEngine	
MakeTransfer_to_DormantAccount_ReturnsError	23 ms	Negative, TransactionEngine	Assert.That(result.IsSuccess, Is.False) Expecte...
MakeTransfer_to_SameAccount_ReturnError	1 ms	Negative, TransactionEngine	
MakeWithdraw_NewDay_MustResetDailyLimit	1 ms	Negative, TransactionEngine	Assert.That(resultSecondWithdrawal.IsSuccess, ...
MakeWithdrawal_ClosedAccount_ReturnsError	< 1 ms	Negative, TransactionEngine	
MakeWithdrawal_ExceedsDailyLimit_ReturnsError	< 1 ms	Negative, TransactionEngine	
MakeWithdrawal_ExceedsMaximumLimit_ReturnsError	< 1 ms	Negative, TransactionEngine	
MakeWithdrawal_InsufficientFunds_ReturnsError	< 1 ms	Negative, TransactionEngine	
MakeWithdrawal_InvalidAccountID_ReturnsError	< 1 ms	Negative, TransactionEngine	
MakeWithdrawal_InvalidAmount_ReturnsError	< 1 ms	Negative, TransactionEngine	
MakeWithdrawal_ValidActiveAccount_UpdatesBalanceAndLogsTransaction	1 ms	Functional, TransactionEngine	
ProvidingNoDate_WhenTryingToRetrieveHistory_ReturnsError	2 ms	Performance, TransactionEn...	
ReverseTransaction_AlreadyReversedTransaction_ReturnsError	5 ms	Negative, TransactionEngine	
ReverseTransaction_FromAccount_WithInsufficientFunds_ReturnsError	1 ms	Negative, TransactionEngine	
ReverseTransaction_InvalidTransactionId("TXN-12345678901234-INVALID", "INVALID REF", "Jaydee V...	< 1 ms	TransactionEngine	
ReverseTransaction_ReversalWindowExpired_ReturnsError	< 1 ms	Negative, TransactionEngine	
ReverseTransaction_ValidTransactionId	1 ms	Positive, TransactionEngine	
TestClosedAccount_ReturnsError	< 1 ms	Negative, TransactionEngine	
TestInvalidAccountID_ReturnsError	< 1 ms	Negative, TransactionEngine	
TryToDeposit_WhenDBCrashes_ThrowsArgException	2 ms	Critical, TransactionEngine	
UsingInvalidAccount_toPull_TransactionHis_ReturnsError	1 ms	Negative, TransactionEngine	

Group Summary

BankCore.Tests.NUnit

Tests in group: 70

Total Duration: 189 ms

Outcomes

68 Passed

2 Failed

Ready

Test Explorer			
Test run finished: 282 Tests (251 Passed, 30 Failed, 1 Skipped) run in 851 ms 0 Warnings 18 Errors			
Test	Dura...	Traits	Error Message
- ApplyForLoanTests (10) - ApplyForLoan_UsingClosedAccount_ReturnsError - ApplyingForALoan_UsingInvalidInfo_ReturnsError (5) - ApplyingForLoan_BUG010 - ApplyingForLoan_SuccessfullyAddsToAuditLog - ApplyingForLoan_UsingInvalidAccount_ReturnsError - AttemptingToApplyForLoan_BelowRequiredCreditScore_ReturnsError - ApproveLoanTests (2) - ApproveForLoan_SuccessfullyAddsToAuditLog - ApprovingValidLoan - CalculateSettlementAmountTests (1) - CalculateSettlement_WrongFeeUsed - CompoundInterestTests (21) - CompoundInterest_AnnualRate_0_Returns0 - CompoundInterest_Calculator_WithDifferentAccTypes (11) - CompoundInterest_InvalidInputs_ThrowsArgException (5) - VariousFreq_Scenarios_CompoundingInterest (4) - DailyInterestTests (10) - DailyInterest_InvalidInputs_ThrowsArxception (4) - DailyInterestTests_MultipleValues_CalculatesCorrectly (5) - DailyIntTest_DuringLeapYear_CalculatesCorrectly - EffectiveAnnualRateTests (6) - EffectiveAnnualRate_InvalidRate_ThrowsException(annualRate: -0,1, Frequency: 12, exceptionMessa... - EffectiveAnnualRate_ValidInputs (5) - FutureValueTests (4) - FutureValue_SimpleIntnt_And_CompoundIntnt_CalculatesCorrectly (2) - FutureValue_SimpleIntnt_And_CompoundIntnt_CalculatesCorrectly(principalAmount: 1000, annualRa... - FutureValue_SimpleIntnt_And_CompoundIntnt_CalculatesCorrectly(principalAmount: 1000, annualRa... - FutureValue_Tests_InvalidPrincipalAmount (2) - GenerateRepaymentScheduleTests (2) - GenerateRepaymentSchedule_MathematicallyCorrect_Over36Months - GenerateRepaymentSchedule_WithInvalidLoanStatus_ReturnsError - GetLoanTests (2) - GetLoan_WithInvalid_Reference - GetLoan_WithValid_Reference - ProcessRepaymentTests (3) - ProcessPayment_BalanceGoesNegative_BUG011 - ProcessPayment_NegativeAmount - ProcessPayment_OnArrearsAccount_ShouldReactivate - RejectLoanTests (2) - SettleLoanTest (1) - SimpleInterestTests (21)	487 ms		
	< 1 ms	Category [Negative]	
	478 ms		
	2 ms	Category [Bug-010]	Expected result.IsSuccess to be False because F...
	7 ms	Category [Positive]	
	< 1 ms	Category [Negative]	
	< 1 ms	Category [Negative]	
	490 ms		
	490 ms	Category [Positive]	
	< 1 ms	Category [Positive]	
	476 ms		
	476 ms	Category [Bug-012]	Expected result.Data to be 18270M, but found...
	378 ms		
	< 1 ms	Category [Positive]	
	377 ms		
	< 1 ms		
	1 ms		
	< 1 ms		
	< 1 ms		
	< 1 ms	Category [Leap Year Test]	
	1 ms		
	1 ms	Category [Negative]	
	< 1 ms		
	467 ms		
	75 ms		
	75 ms	Category [Positive]	Expected result to approximate 1120M +/- 0.01...
	< 1 ms	Category [Positive]	
	392 ms		
	485 ms		
	485 ms	Category [Positive]	
	< 1 ms	Category [Negative]	
	478 ms		
	477 ms	Category [Negative]	
	1 ms	Category [Positive]	
	485 ms		
	2 ms	Category [Bug-011]	Expected result.Data.OutstandingBalance to be...
	476 ms	Category [Negative]	
	7 ms	Category [Positive]	
	489 ms		
	484 ms		
	69 ms		

- SimpleInterestTests (21) - SimpleInterest_InvalidInputs (3) - SimpleInterest_ReturnsError_WhenMonthsIsNegative - SimpleInterest_ReturnsError_WhenPrincipalIsNegative - SimpleInterest_ReturnsError_WhenRateIsNegative - SimpleInterest_UsingClassData_CalculatesCorrectly(principal: 1000, rate: 0,05, months: 12, expected:... - SimpleInterest_VariousCombinations_ReturnsEachExpected (11) - SimpleInterest_VariousCombinations_ReturnsEachExpected(principalAmount: 1000, annualRate: 0,... - SimpleInterest_VariousCombinations_ReturnsEachExpected(principalAmount: 10000, annualRate:... - SimpleInterest_VariousCombinations_ReturnsEachExpected(principalAmount: 10000, annualRate:... - SimpleInterest_VariousCombinations_ReturnsEachExpected(principalAmount: 10000, annualRate:... - SimpleInterest_VariousCombinations_ReturnsEachExpected(principalAmount: 100000, annualRate:... - SimpleInterest_VariousCombinations_ReturnsEachExpected(principalAmount: 100000, annualRate:... - SimpleInterest_VariousCombinations_ReturnsEachExpected(principalAmount: 100000, annualRate: 0,001, periodInMonths:... - SimpleInterest_VariousCombinations_ReturnsEachExpected(principalAmount: 12000, annualRate:... - SimpleInterest_VariousCombinations_ReturnsEachExpected(principalAmount: 250, annualRate: 0,0... - SimpleInterest_VariousCombinations_ReturnsEachExpected(principalAmount: 35000, annualRate:... - SimpleInterest_VariousCombinations_ReturnsEachExpected(principalAmount: 365, annualRate: 0,0... - SimpleInterest_VariousCombinations_ReturnsEachExpected(principalAmount: 5000, annualRate: 0,... - SimpleInterestCalculation_With0PercentAnnualRate - SimpleInterestTest_11PercentInterest_OverOneYear - SimpleInterestCalculation_With0Months_ShouldReturn0	69 ms		
	< 1 ms		
	1 ms	Category [Negative]	
	< 1 ms	Category [Negative]	
	< 1 ms	Category [Negative]	
	< 1 ms	Category [Positive]	Expected result to approximate 50.00M +/- 0.0...
	68 ms		
	< 1 ms	Category [Positive]	Expected result to approximate 100M +/- 0.01...
	< 1 ms	Category [Positive]	Expected result to approximate 1M +/- 0.01M...
	< 1 ms	Category [Positive]	Expected result to approximate 500M +/- 0.01...
	1 ms	Category [Positive]	Expected result to approximate 10000M +/- 0.0...
	< 1 ms	Category [Positive]	Expected result to approximate 100M +/- 0.01...
	< 1 ms	Category [Positive]	Expected result to approximate 1000M +/- 0.01...
	< 1 ms	Category [Positive]	Expected result to approximate 375M +/- 0.01...
	< 1 ms	Category [Positive]	Expected result to approximate 30M +/- 0.01M...
	< 1 ms	Category [Positive]	Expected result to approximate 4200M +/- 0.01...
	< 1 ms	Category [Positive]	Expected result to approximate 2.43M +/- 0.01...
	67 ms	Category [Positive]	Expected result to approximate 150M +/- 0.01...
	< 1 ms	Category [Positive]	
	< 1 ms	Category [Positive]	Expected result to approximate 1650.00M +/- 0...
	< 1 ms	Category [Negative]	System.ArgumentException : Months must be p...